# Lab 07: Network Protocols

Operating Systems and Distributed Systems

The purpose of this lab is to introduce tools that provide valuable information when working with networks. Additionally, it enforces the students' understanding of the layers in both the OSI and TCP/IP models.

The lab is structured so the student is assisted through the tasks.

Remember to read the exercise and the accompanying text thoroughly before asking for help.

## Contents

# Exercise 01: Examine the tools

In this exercise, the student is going to use the Linux tools `ip` and `netstat`.

The tools are used to configure network interfaces, manipulate the routing table and show network status.

`$ ip link` is used to show the devices interfaces.

`$ ip address` is used to show the ip's that the device's interfaces hold.

`$ ip route` is used to show the networking routes of the device.

`$ netstat` is used to show current network connections.

### Your task

1. Examine the commands `ip link`, `ip address`, `ip route` and `netstat` and document their purposes and how they work.

The exercise will return the following message:

```
$ ip link
1: wlp0s20f3: <BROADCAST,MULTICAST,UP,LOWER_UP> state UP group default
    link/ether a3:12:03:ff:6c:c3 brd ff:ff:ff:ff:ff:ff
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> state UP group default
    link/ether ff:e0:4e:6a:ff:d5 brd ff:ff:ff:ff:ff:ff


$ ip address
1: wlp0s20f3: <BROADCAST,MULTICAST,UP,LOWER_UP> state UP group default
    link/ether a3:12:03:ff:6c:c3 brd ff:ff:ff:ff:ff:ff
    inet 10.94.100.100/24 brd 10.94.127.255 scope global wlp0s20f3
    inet6 fe80::7bed:a5b1:82d5:91ae/64 scope link noprefixroute
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP>  state UP group default
    link/ether ff:e0:4e:6a:ff:d5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.11/24 scope global enx00e04e6a27d5
      valid_lft forever preferred_lft forever


$ ip route
default via 10.94.100.1 dev wlp0s20f3 proto dhcp metric 600
10.94.100.0/19 dev wlp0s20f3 kernel scope link src 10.94.100.100 metric 600
```

```
192.168.10.0/24 dev eth0 proto kernel scope link src 192.168.10.11

$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0     64 192.168.10.11:ssh       192.168.10.10:56600     ESTABLISHED
```

1. ip link
Purpose: Shows the network interfaces configured on the system.
Syntax: ip link [show|set|add|delete]
Example: `ip link show` displays the status of all interfaces.

2. ip address
Purpose: Displays or configures the IP addresses assigned to network interfaces.
Example: `ip address show eth0` shows information only for the eth0 interface.

3. ip route
Purpose: Displays or modifies the routing table.
Example: ip route add 192.168.1.0/24 via 192.168.0.1 adds a new route.

4. netstat
Purpose: Shows current network connections and open ports.
Example: `netstat -tuln` lists all TCP and UDP ports currently listening.

## Exercise 02: Take your own route

In this exercise, two or more people are going to connect their computers either directly, through a switch or a mobile hotspot.

**Important 1**

Disable automatic IP assignment in your network settings.

**Hint 1**

Assignment of IP addresses to network interfaces can either be done through the GUI or tools like `ip`.

**Hint 2**

Creation of routes can be done using the `ip` tool.

### Your task

1. Connect your computer to another computer, either directly, through a switch or a mobile hotspot
2. Assign an IP address to your network interface sudo ip address add 192.168.10.2/24 dev eth0 (to do for each PC, different IP) ip address show eth0 (to check)
3. If no route is created, then a route entry has to be created to enable communication with the new network sudo ip route add 192.168.10.0/24 dev eth0 ip route (to check)
4. Spin up the Docker Compose exercise from Lab 02
5. Let someone connected to your computer (either directly or indirectly) visit your Docker Compose website through your selected IP address http://192.168.10.2:8080

# Exercise 03: Networking Models

In this exercise, the student is going to investigate the two network models (OSI and TCP/IP), explain how they work and what happens at each layer.

### Your task

1. Discuss the differences between the OSI model and the TCP/IP model
2. Which layers in the OSI model correspond to which layers in the TCP/IP model
3. What is the responsibility of each layer?
4. What does it add or remove?
5. How does addressing work on the layer? Do you know some details about how the layer is implemented?

1. Differences between OSI and TCP/IP models
The OSI model (Open Systems Interconnection) is a theoretical framework with seven layers, used for understanding how data communication works.
The TCP/IP model is a practical implementation used in real networks and has four layers.
OSI separates functions more precisely (presentation, session, etc.), while TCP/IP combines them for efficiency.
OSI is mainly used for teaching and conceptual understanding; TCP/IP is used in real-world networking.

| **OSI Model (7 Layers)** | **TCP/IP Model (4 Layers)** |
| ------------------------ | --------------------------- |
| 7. Application | Application |
| 6. Presentation | Application |
| 5. Session | Application |
| 4. Transport | Transport |
| 3. Network | Internet |
| 2. Data Link | Network Access |
| 1. Physical | Network Access |

3. Responsibility of each layer

Application (OSI 7 / TCP-IP 4): Provides network services to end-users (HTTP, FTP, SMTP, DNS).

Presentation (OSI 6): Translates, encrypts, or compresses data formats (e.g., ASCII, JPEG).

Session (OSI 5): Manages communication sessions (start, maintain, terminate).

Transport (OSI 4 / TCP-IP 3): Ensures reliable delivery and flow control (TCP, UDP).

Network (OSI 3 / TCP-IP 2): Determines routing and logical addressing (IP, ICMP).

Data Link (OSI 2 / TCP-IP 1): Handles frames and physical addressing (MAC addresses, Ethernet).

Physical (OSI 1 / TCP-IP 1): Manages transmission of bits through cables or wireless media.

4. What each layer adds or removes
Each layer adds a header containing control information when data moves down the stack (encapsulation).
When data is received, each layer removes its header (decapsulation).
Example:
Transport layer adds a TCP header (ports).
Network layer adds an IP header (IP address).
Data Link layer adds MAC header/trailer (frame).

| **Layer** | **Address Type** | **Example / Implementation** |
| ----------- | ------------------------- | --------------------------------------------- |
| Application | Logical (domain name) | [www.example.com](http://www.example.com), DNS |
| Transport | Port numbers | 80 (HTTP), 443 (HTTPS) |
| Network | IP address | 192.168.1.10 |
| Data Link | MAC address | 00:1A:2B:3C:4D:5E |
| Physical | No address (uses signals) | Ethernet cables, Wi-Fi radio |

Each layer uses its own addressing scheme to identify devices or processes.For example, the IP layer handles logical routing between networks, while the MAC layer identifies physical devices on the same network.

## Exercise 04: To Send a Package

In this exercise, the student is going to explain how packages are sent over a network.

### Your task

1. Explain step by step how a package from Computer A with the ip `185.8.135.136` gets to Computer B with the ip `101.24.34.2`
2. Which layers in the TCP/IP model does communication happen through?

1. Step-by-step: how a packet goes from 185.8.135.136 (A) to 101.24.34.2 (B)1. Application (on A):An app prepares data for B's I

# Exercise 05: NAT

In this exercise, the student is going to explain what Network Address Translation (NAT) is and how it works. You may have to research a little to answer the questions.

**Your task**

1. Explain what NAT is
2. How does NAT work?
3. When is NAT useful?

Report – Exercise 051. What is NAT?Network Address Translation (NAT) is a process that allows a router or firewall to modify th